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650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
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1 Scope

The present specification provides the stage 3 definition of the Binding Support Management Service of the 5G System.

The 5G System Architecture is defined in 3GPP TS 23.501 [2]. The stage 2 definition and related procedures for Binding Support Management Service is specified in 3GPP TS 23.502 [3] and 3GPP TS 23.503 [4].

The 5G System stage 3 call flows are provided in 3GPP TS 29.513 [5].

The Technical Realization of the Service Based Architecture and the Principles and Guidelines for Services Definition are specified in 3GPP TS 29.500 [6] and 3GPP TS 29.501 [7].

The Binding Support Management Service is provided by the Binding Support Function (BSF).

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".

[3] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[4] 3GPP TS 23.503: "Policy and Charging Control Framework for the 5G System; Stage 2".

[5] 3GPP TS 29.513: "5G System; Policy and Charging Control signalling flows and QoS parameter mapping; Stage 3".

[6] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".

[7] 3GPP TS 29.501: "5G System; Principles and Guidelines for Services Definition; Stage 3".

[8] IETF RFC 7540: "Hypertext Transfer Protocol Version 2 (HTTP/2)".

[9] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

[10] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces Stage 3".

[11] OpenAPI: "OpenAPI 3.0.0 Specification", https://github.com/OAI/OpenAPI-Specification/blob/master/versions/3.0.0.md.

[12] 3GPP TS 29.510: "5G System; Network Function Repository Services; Stage 3".

[13] IETF RFC 7807: "Problem Details for HTTP APIs".

[14] 3GPP TS 29.213: " Policy and Charging Control signalling flows and Quality of Service (QoS) parameter mapping".

[15] 3GPP TS 33.501: "Security architecture and procedures for 5G system".

[16] IETF RFC 6749: "The OAuth 2.0 Authorization Framework".

[17] 3GPP TS 23.527: "5G System; Restoration Procedures".

[18] 3GPP TR 21.900: "Technical Specification Group working methods".

[19] 3GPP TS 23.316: "Wireless and wireline convergence access support for the 5G System (5GS)".

[20] IETF RFC 7396: "JSON Merge Patch".

[21] 3GPP TS 29.512: "5G System; Session Management Policy Control Service; Stage 3".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

5G-RG	5G Residential Gateway
AF	Application Function
BSF	Binding Support Function
DNN	Data Network Name
DRA	Diameter Routing Agent
HTTP	Hypertext Transfer Protocol
FN-RG	Fixed Network Residential Gateway
FQDN	Fully Qualified Domain Name
GPSI	Generic Public Subscription Identifier
JSON	JavaScript Object Notation HTTP Hypertext Transfer Protocol
MAC	Media Access Control
NEF	Network Exposure Function
NRF	Network Repository Function
NWDAF	Network Data Analytics Function
PCF	Policy Control Function
SMF	Session Management Function
S-NSSAI	Single Network Slice Selection Assistance Information
SUPI	Subscription Permanent Identifier
UDR	Unified Data Repository

4 Binding Support Management Service

4.1 Service Description

4.1.1 Overview

The Binding Support Management Service as defined in 3GPP TS 23.502 [3] and 3GPP TS 23.503 [4], is provided by the Binding Support Function (BSF).

The Nbsf_Management service is used to provide a PDU session binding functionality, which ensures that an AF request for a certain PDU Session reaches the relevant PCF holding that PDU Session information, or ensures that the same PCF is selected for multiple PDU sessions.

This service:

- allows NF service consumers to register, update and remove binding information; and
- allows NF service consumers to retrieve binding information.

4.1.2 Service Architecture

The 5G System Architecture is defined in 3GPP TS 23.501 [2]. The Policy and Charging related 5G architecture is also described in 3GPP TS 23.503 [4] and 3GPP TS 29.513 [5].

The Binding Support Management Service (Nbsf_Management) is exhibited by the Binding Support Function (BSF).

The known consumers of the Nbsf_Management service are:

- Policy Control Function (PCF)
- Network Exposure Function (NEF)
- Application Function (AF); and
- Network Data Analytics Function (NWDAF)

As described in 3GPP TS 23.503 [4], the BSF is a function that can be deployed standalone or as a functionality provided by other network functions, such as PCF, UDR, NRF, SMF.

NOTE 1: The PCF accesses the Nbsf_Management service at the BSF via an internal interface when it is collocated with BSF.

NOTE 2: The DRA decides to select a BSF based on user IP address range when the DRA has no binding information for the subscriber to get the relevant PCF address. DRA and BSF coexistence is described in 3GPP TS 29.513 [5], Annex A.

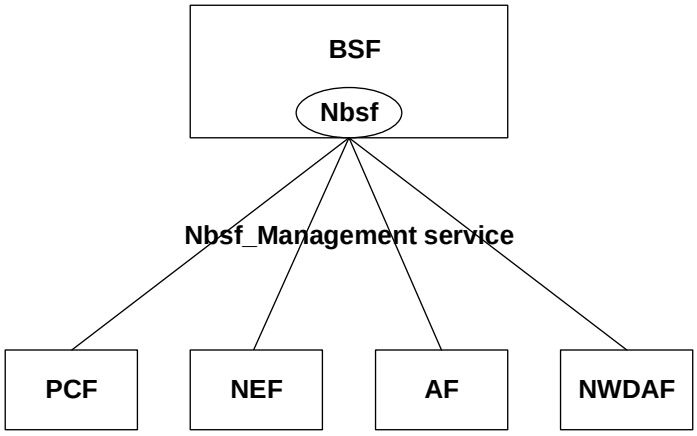


Figure 4.1.2-1: Reference Architecture for the Nbsf_Management service; SBI representation

4.1.3 Network Functions

4.1.3.1 Binding Support Function (BSF)

The BSF:

- stores the binding information for a certain PDU Session; and
- enables the discovery of binding information (e.g. the address information of the selected PCF).

The BSF allows NF service consumers (e.g. PCF) to register, update and remove a binding information, and allows NF service consumers (e.g. AF, NEF, NWDAF) to discover a binding information (e.g. the address information of the selected PCF).

The BSF can be deployed standalone or collocated with other network functions, such as PCF, UDR, NRF and SMF.

4.1.3.2 NF Service Consumers

The Policy Control Function (PCF):

- registers binding information in the BSF for a UE when an IPv4 address and/or IPv6 prefix is allocated, or a MAC address is used for the PDU session;
- updates binding information in the BSF when a UE address information is changed for the PDU Session; and
- removes binding information in the BSF when an IPv4 address and/or IPv6 prefix is released, or a MAC address is not used for the PDU Session.

The Network Exposure Function (NEF):

- provides means for the Application Functions to securely interact with the Policy framework for policy control to 3GPP network. During the procedure, it needs to discover the selected PCF by using the Nbsf_Management_Discovery service operation.

The Application Function (AF):

- discovers the selected PCF by using the Nbsf_Management_Discovery service operation when it is allowed to interact directly with the policy framework for policy control.

The Network Data Analytics Function (NWDAF):

- discovers the selected PCF by using the Nbsf_Management_Discovery service operation.

4.2 Service Operations

4.2.1 Introduction

Table 4.2.1-1: Operations of the Nbsf_Management Service

Service operation name	Description	Initiated by
Nbsf_Management_Register	This service operation is used to register the binding information for a UE.	NF service consumer (PCF)
Nbsf_Management_Deregister	This service operation is used to deregister the binding information for a UE.	NF service consumer (PCF)
Nbsf_Management_Discovery	This service operation is used by an NEF or AF or NWDAF to discover a selected PCF.	NF service consumer (NEF, AF, NWDAF)
Nbsf_Management_Update	This service operation is used to update an existing session binding information for a UE.	NF service consumer (PCF)

4.2.2 Nbsf_Management_Register Service Operation

4.2.2.1 General

This service operation allows a NF service consumer (e.g. PCF) to register the session binding information for a UE in the BSF by providing the user identity, the DNN, the UE address(es) and the selected PCF address for a certain PDU Session to the BSF, and BSF stores the information.

If the BindingUpdate feature is not supported and the NF service consumer (e.g. PCF) receives a new UE address (e.g. IPv6 prefix) and has already registered session binding information for this PDU session, the NF service consumer (e.g. PCF) shall register a new session binding information in the BSF.

If the SamePcf feature or the ExtendedSamePcf feature is supported, this service operation allows the PCF to check whether PCF addressing information for Npcf_SMPolicyControl service is already registered in the BSF by another PCF for a combination of the UE ID, DNN and S-NSSAI parameters of the PDU session.

4.2.2.2 Register a new PCF Session binding information

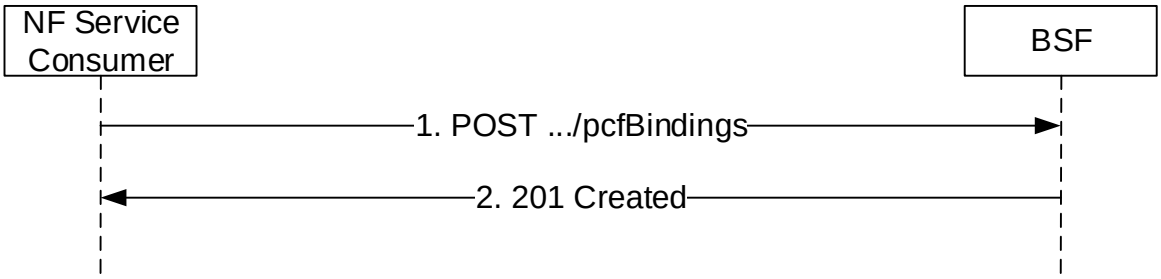


Figure 4.2.2.2-1: NF service consumer register a new PCF Session binding information

The NF service consumer shall invoke the Nbsf_Management_Register service operation to register the session binding information for a UE in the BSF. The NF service consumer shall send for this an HTTP POST request with "{apiRoot}/nbsf-management/v1/pcfBindings" as Resource URI representing the "PCF Session Bindings", as shown in figure 4.2.2.2-1, step 1, to create a binding information for an "Individual PCF Session Binding" according to the information (e.g. UE address(es), SUPI, GPSI, DNN, S-NSSAI) in the message body. When the "ExtendedSamePcf" feature is not supported, the "PcfBinding" data structure provided in the request body shall include:

- if the "MultiUeAddr" feature is not supported or not yet known, address information of the served UE consisting of:

  (i) either IP address information consisting of:

  + the IPv4 address encoded as "ipv4Addr" attribute; and/or

  + the /128 IPv6 address, the IPv6 address prefix or an IPv6 prefix shorter than /64 encoded as "ipv6Prefix" attribute; or

  (ii) the MAC address encoded as "macAddr48" attribute;

  Otherwise, address information of the served UE consisting of:

  (i) any IP address information consisting of:

  + the IPv4 address encoded as "ipv4Addr" attribute;

  + the /128 IPv6 address, the IPv6 address prefix or an IPv6 prefix shorter than /64 encoded as "ipv6Prefix" attribute; and/or

  + the additional /128 IPv6 addresses, the IPv6 address prefixes or IPv6 prefixes shorter than /64 encoded as "addIpv6Prefixes" attribute; or

  (ii) the MAC address encoded as "macAddr48" attribute and/or the additional MAC addresses encoded as "addMacAddrs" attribute;

- PCF address information consisting of:

  (i) if the PCF supports the Npcf_PolicyAuthorization service:

  + the FQDN of the PCF encoded as "pcfFqdn" attribute; and/or

  + a description of IP endpoints at the PCF hosting the Npcf_PolicyAuthorization service encoded as "pcfIpEndPoints" attribute; and

  (ii) if the PCF supports the Rx interface:

  + the Diameter host id of the PCF encoded as "pcfDiamHost"; and

  + the Diameter realm of the PCF encoded as"pcfDiamRealm" attributes;

- DNN encoded as "dnn" attribute;

- S-NSSAI encoded as "snssai" attribute; and

- if the "SamePcf" feature defined in subclause 5.8 is supported and the PCF determines based on operator policies that the same PCF shall be selected for the SM Policy associations:

  (i) PCF address information for Npcf_SMPolicyControl service consisting of:

  + the FQDN of the PCF encoded as "pcfSmFqdn" attribute; or

  + a description of IP endpoints at the PCF hosting the Npcf_SMPolicyControl service encoded as "pcfSmIpEndPoints" attribute; and

  (ii) the parameters combination for selecting the same PCF encoded within the "paraCom" attribute if the PCF registers the binding information for the indicated parameter combination for the first time.

NOTE 1: When the "SamePcf" feature is supported, the PCF omits the "paraCom" attribute when creates the corresponding binding information related to the subsequent PDU sessions for the same parameter combination.

and may include:

- SUPI encoded as "supi" attribute;
- GPSI encoded as "gpsi" attribute;
- IPv4 address domain encoded as "ipDomain" attribute; and
- framed routes consisting of:

  (i) one or more framed routes within the "ipv4FrameRouteList" attribute for IPv4; and/or

  (ii) one or more framed routes within the "ipv6FrameRouteList" attribute for IPv6.

When the "TimeSensitiveNetworking" feature is supported by the PCF as defined in subclause 5.8 of 3GPP TS 29.512 [21], the address information of the served UE contains the MAC address of the DS-TT encoded in the "macAddr48" attribute as received by the PCF when reporting the bridge information attribute.

When the "ExtendedSamePcf" feature is supported the address information of the served UE may be provided if available, i.e., the "ipv4Addr", the "ipv6Prefix" and/or "addIpv6Prefixes" attributes or the "macAddr48" and/or "addMacAddrs" attributes may be provided if available.

When the "ExtendedSamePcf" feature is supported the PCF address for the Npcf_PolicyAuthorization and/or Rx interface may be provided if available, i.e., the "pcfFqdn" and/or the "pcfIpEndPoints" attributes, and/or the "pcfDiamHost" and/or the "pcfDiamRealm" attributes may be provided if available.

NOTE 2: Before requesting the BSF to check if there is an existing PCF binding information for the same UE ID, S-NSSAI and DNN combination registered by other PCF(s), the PCF determines whether the BSF supports the "SamePcf" and/or "ExtendedSamePcf" features either via local configuration or by checking the BSF profile retrieved from the NRF as specified in 3GPP TS 29.510 [12].

Upon the reception of an HTTP POST request with: "{apiRoot}/nbsf-management/v1/pcfBindings" as Resource URI and "PcfBinding" data structure as request body, the BSF shall:

- create new binding information;
- assign a bindingId; and
- store the binding information.

The PCF as NF service consumer may provide PCF Id in "pcfId" attribute and recovery timestamp in "recoveryTime" attribute. The BSF may use the "pcfId" attribute to supervise the status of the PCF as described in subclause 5.2 of 3GPP TS 29.510 [12] and perform necessary clean up upon status change of the PCF later, and/or both the "pcfId" attribute and the "recoveryTime" attribute in clean up procedure as described in subclause 6.4 of 3GPP TS 23.527 [17].

The PCF as a NF service consumer may provide PCF Set Id within the "pcfSetId" attribute and "bindLevel" attribute set to NF_SET or provide PCF Set Id within the "pcfSetId" attribute, PCF instance Id within the "pcfId" attribute and "bindLevel" attribute set to NF_INSTANCE.

If the BSF created an "Individual PCF Session Binding" resource, the BSF shall respond with "201 Created" status code with the message body containing a representation of the created binding information, as shown in figure 4.2.2.2-1, step 2. The BSF shall include a Location HTTP header field containing the URI of the created binding information, i.e. "{apiRoot}/nbsf-management/v1/pcfBindings/{bindingId}".

If errors occur when processing the HTTP POST request, the PCF shall apply error handling procedures as specified in subclause 5.7.

If the "SamePcf" feature defined in subclause 5.8 is supported and the "paraCom" attribute is included in the HTTP POST message, the BSF shall check the received "paraCom" attribute. If the BSF detects that there is an existing PCF binding information including the same "dnn", "snssai" and "supi" attribute values as each of the corresponding attribute values within the "paraCom" attribute, the BSF shall reject the request with an HTTP "403 Forbidden" status code and shall include in the response the "ExtProblemDetails" data structure including the FQDN of the existing PCF hosting the Npcf_SMPolicyControl service within the "pcfSmFqdn" attribute or the description of IP endpoints at the existing PCF hosting the Npcf_SMPolicyControl service within the "pcfSmIpEndPoints" attribute of "BindingResp" data structure, and the "cause" attribute of the "ProblemDetails" data structure set to "EXISTING_BINDING_INFO_FOUND".

4.2.3 Nbsf_Management_Deregister Service Operation

4.2.3.1 General

This service operation allows the service consumer to remove the session binding information for a UE in the BSF. It is executed by deleting a given resource identified by an "Individual PCF Session Binding" resource identifier. The operation is invoked by issuing an HTTP DELETE request on the URI representing the specific session binding information.

4.2.3.2 Deregister an individual PCF Session binding information

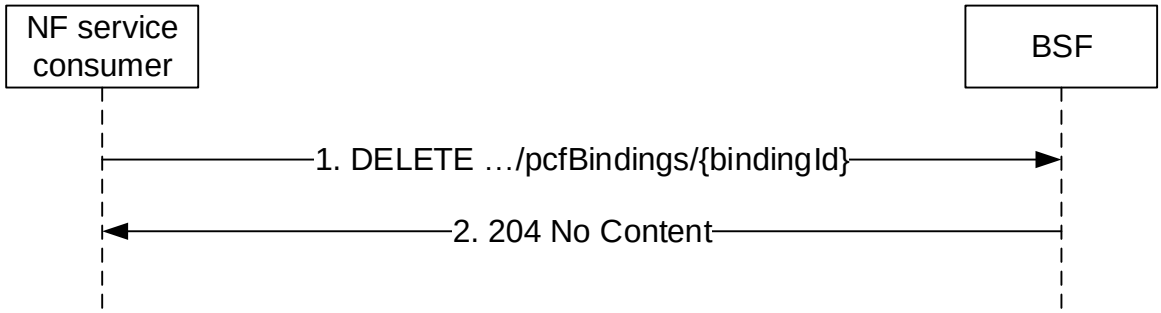


Figure 4.2.3.2-1: Session Binding Information Deregistration

The NF service consumer shall invoke the Nbsf_Management_Deregister service operation to deregister the session binding information for a UE in the BSF. The NF service consumer shall send an HTTP DELETE request with "{apiRoot}/nbsf-management/v1/pcfBindings/{bindingId}" as Resource URI, where "{bindingId}" is the "Individual PCF Session Binding" resource identifier that is to be deleted, as shown in figure 4.2.3.2-1, step 1.

Upon the reception of an HTTP DELETE request with: "{apiRoot}/nbsf-management/v1/pcfBindings/{bindingId}" as Resource URI, the BSF shall:

- remove the corresponding binding information.

If the HTTP DELETE request message from the NF service consumer is accepted, the BSF shall respond with "204 No Content" status code, as shown in figure 4.2.3.2-1, step 2.

If errors occur when processing the HTTP DELETE request, the BSF shall send an HTTP error response as specified in subclause 5.7.

If the Individual PCF Session Binding resource does not exist, the BSF shall respond with "404 Not Found" error code.

If the feature "ES3XX" is supported, and the BSF determines the received HTTP DELETE request needs to be redirected, the BSF shall send an HTTP redirect response as specified in subclause 6.10.9 of 3GPP TS 29.500 [6].

4.2.4 Nbsf_Management_Discovery Service Operation

4.2.4.1 General

This service operation allows the service consumer to use the HTTP GET method to obtain the address information of the selected PCF.

4.2.4.2 Retrieve the PCF Session binding information for a given tuple

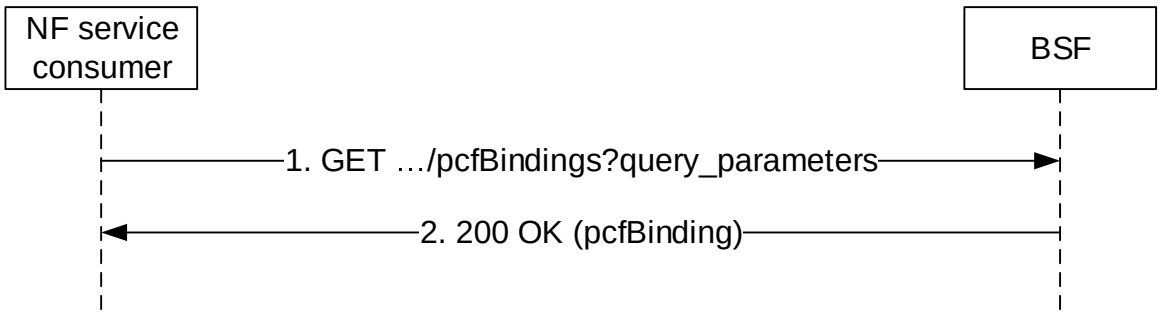


Figure 4.2.4.2-1: NF service consumer retrieve the PCF Session binding information for a given tuple

The NF service consumer shall invoke the Nbsf_Management_Discovery service operation to obtain address information of the selected PCF for a PDU session in the BSF. The NF service consumer shall send an HTTP GET request with "{apiRoot}/nbsf-management/v1/pcfBindings" as Resource URI, and "query parameters" that shall include:

- UE address;

and may include:

- SUPI or GPSI;
- DNN and optionally S-NSSAI; and
- IPv4 address domain.

NOTE: The query parameters S-NSSAI and/or IPv4 address domain are helpful in the scenario of IPv4 address overlapping where the same IPv4 address may be allocated to UE PDU sessions.

Upon the reception of an HTTP GET request with: "{apiRoot}/nbsf-management/v1/pcfBindings" as Resource URI, the BSF shall search the corresponding binding information. If "ipv6Prefix" is used as an UE IPv6 address in the query parameters, the BSF shall use the longest prefix match to find a matching IPv6 prefix so that the IPv6 address in the query parameters is within the address range covered by that matching IPv6 prefix. The IPv6 address in the query parameters shall be formatted as an IPv6 prefix value including the trailing prefix length "/128". If the framed routes exist in the binding information, the BSF shall use framed routes to match the UE address in the query parameters.

If the HTTP request message from the NF service consumer is accepted and a session binding resource matching the query parameters exists, the BSF shall reply with an HTTP "200 OK" response, as shown in figure 4.2.4.2-1, step 2, containing the corresponding "PcfBinding" data structure, as provided by the PCF during the Nbsf_Management_Register Service Operation, in the response body containing PCF addressing information, and if available, the related PCF Set Id and PCF instance Id. If there is no PCF session binding information matching the query parameters, the BSF shall respond with an HTTP "204 No Content".

NOTE 2: If the NF service consumer (such as the AF or NEF) is not able to reach the received PCF address(es), the NF service consumer can use the PCF Set Id and the PCF instance Id as specified in 3GPP TS 29.513 [5] subclause 6.2.

If the "PCF Session Bindings" resource does not exist, the BSF shall respond with "404 Not Found" HTTP error code. If an invalid combination of query parameters (i.e. a combination without UE address) is contained in the request URI, the BSF shall respond with an HTTP "400 Bad Request" error code containing "MANDATORY_QUERY_PARAM_MISSING" as application error within the ProblemDetails IE. If more than one

PCF Session Binding resources are found, the BSF shall respond with an HTTP "400 Bad Request" error code containing "MULTIPLE_BINDING_INFO_FOUND" as application error within the ProblemDetails IE.

4.2.5 Nbsf_Management_Update Service Operation

4.2.5.1 General

This service operation allows the NF service consumer to update an existing session binding information for a UE in the BSF by providing information to be updated (e.g. the UE address(es)) for a PDU Session, and BSF updates the session binding information.

4.2.5.2 Update an existing PCF Session binding information

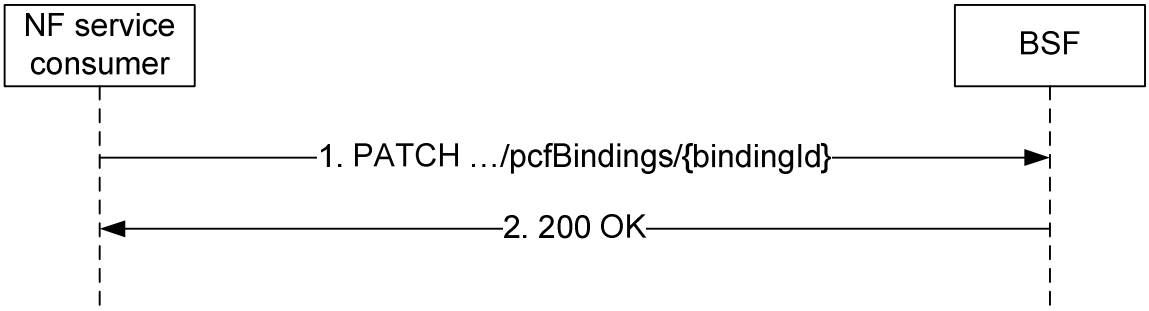


Figure 4.2.5.2-1: NF service consumer update an existing PCF Session binding information

If the feature "BindingUpdate" is supported, the NF service consumer shall invoke the Nbsf_Management_Update service operation to update the session binding information for a UE in the BSF. The NF service consumer shall send an HTTP PATCH request with "{apiRoot}/nbsf-management/v1/pcfBindings/{bindingId}" as Resource URI, where "{bindingId}" is the "Individual PCF Session Binding" resource identifier that is to be updated, as shown in figure 4.2.5.2-1, step 1. The "PcfBindingPatch" data structure provided in the request body shall contain the information to be updated as follows.

The "PcfBindingPatch" data structure:

- for the IP address information of the served UE:

  a) shall contain the "ipv4Addr" attribute if the IPv4 address is modified, or if the "ExtendedSamePcf" feature is supported, if the IPv4 address was not previously provided, and may contain the "ipDomain" attribute if the IPv4 address domain is modified or if the "ExtendedSamePcf" feature is supported, if the IPv4 address domain was not previously provided and applies. To remove the IPv4 address the "ipv4Addr" attribute shall be set to "null" and if applicable, the "ipDomain" attribute shall be set to "null"; and/or

  b) shall contain the "ipv6Prefix" attribute if the IPv6 address information is modified, or if the "ExtendedSamePcf" feature is supported, if the IPv6 address information was not previously provided. The "ipv6Prefix" attribute shall be set to "null" if the IPv6 address information is removed; and/or

  c) if the "MultiUeAddr" feature is supported, shall contain:

     1) the "addIpv6Prefixes" attribute containing the new complete list of additional IPv6 Address Prefixes if the additional IPv6 address information is modified, or if the "ExtendedSamePcf" feature is supported, the current list of IPv6 address prefixes if it was not previously provided; or

     2) the "addIpv6Prefixes" attribute set to "null" if all additional IPv6 Address Prefixes are removed; or

- for the MAC address information of the served UE:

  a) shall contain the "macAddr48" attribute if the MAC address is modified, or if the "ExtendedSamePcf" feature is supported, if the MAC address was not previously provided. The "macAddr48" attribute shall be set to "null" if the MAC address is removed; and/or

  b) if the "MultiUeAddr" feature is supported, shall contain:

1) the "addMacAddrs" attribute containing the new complete list of additional MAC addresses if the additional MAC address information is modified, or if the "ExtendedSamePcf" feature is supported, the current list of MAC address(es) if it was not previously provided; or

2) the "addMacAddrs" attribute set to "null" if all additional MAC addresses are removed; or

- for the PCF instance and the associated PCF address information of the PCF holding the SM policy association, should contain if a new PCF instance is selected:

a) the PCF instance ID encoded as "pcfId" attribute;

b) if the PCF supports the Npcf_PolicyAuthorization service:

1) the FQDN of the PCF encoded as "pcfFqdn" attribute; and/or

2) a description of IP endpoints at the PCF hosting the Npcf_PolicyAuthorization service encoded as "pcfIpEndPoints" attribute; and/or

c) if the PCF supports the Rx interface:

1) the Diameter host id of the PCF encoded as "pcfDiamHost"; and

2) the Diameter realm of the PCF and "pcfDiamRealm" attributes.

If the BSF cannot successfully fulfil the received HTTP PATCH request due to the internal BSF error or due to the error in the HTTP PATCH request, the BSF shall send the HTTP error response as specified in subclause 5.7.

Otherwise, upon the reception of the HTTP PATCH request with: "{apiRoot}/nbsf-management/v1/pcfBindings/{bindingId}" as Resource URI and the "PcfBindingPatch" data structure as request body, the BSF shall update the binding information.

If the BSF successfully updated an "Individual PCF Session Binding" resource, the BSF shall respond with "200 OK" status code with the message body containing the resource representation with the updated session binding information in the "PcfBinding" data structure, as shown in figure 4.2.5.2-1, step 2.

If errors occur when processing the HTTP PATCH request, the BSF shall send an HTTP error response as specified in subclause 5.7.

If the feature "ES3XX" is supported, and the BSF determines the received HTTP PATCH request needs to be redirected, the BSF shall send an HTTP redirect response as specified in subclause 6.10.9 of 3GPP TS 29.500 [6].

5 Nbsf_Management Service API

5.1 Introduction

The Nbsf_Management Service shall use the Nbsf_Management API.

The API URI of the Nbsf_Management API shall be:

{apiRoot}/<apiName>/<apiVersion>/

The request URIs used in HTTP requests from the NF service consumer towards the BSF shall have the Resource URI structure defined in subclause 4.4.1 of 3GPP TS 29.501 [7], i.e.:

{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>

with the following components:

- The {apiRoot} shall be set as described in 3GPP TS 29.501 [7].
- The <apiName> shall be "nbsf-management".
- The <apiVersion> shall be "v1".

- The <apiSpecificResourceUriPart> shall be set as described in subclause 5.3.

5.2 Usage of HTTP

5.2.1 General

HTTP/2, IETF RFC 7540 [8], shall be used as specified in clause 5 of 3GPP TS 29.500 [6].

HTTP/2 shall be transported as specified in subclause 5.3 of 3GPP TS 29.500 [6].

The OpenAPI [11] specification of HTTP messages and content bodies for the Nbsf_Management is contained in Annex A.

5.2.2 HTTP standard headers

5.2.2.1 General

See subclause 5.2.2 of 3GPP TS 29.500 [6] for the usage of HTTP standard headers.

5.2.2.2 Content type

JSON, IETF RFC 8259 [9], shall be used as content type of the HTTP bodies specified in the present specification as specified in subclause 5.4 of 3GPP TS 29.500 [6]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 7807 [13].

JSON object used in the HTTP PATCH request shall be encoded according to "JSON Merge Patch" and shall be signalled by the content type "application/merge-patch+json", as defined in IETF RFC 7396 [20].

5.2.3 HTTP custom headers

5.2.3.1 General

The Nbsf_Management Service API shall support HTTP custom header fields specified in subclause 5.2.3.2 of 3GPP TS 29.500 [6].

In this release of the specification, no specific custom headers are defined for the Nbsf_Management Service API.

5.3 Resources

5.3.1 Resource Structure

The structure of the Resource URI of the Nbsf_Management service is shown in figure 5.3.1-1.

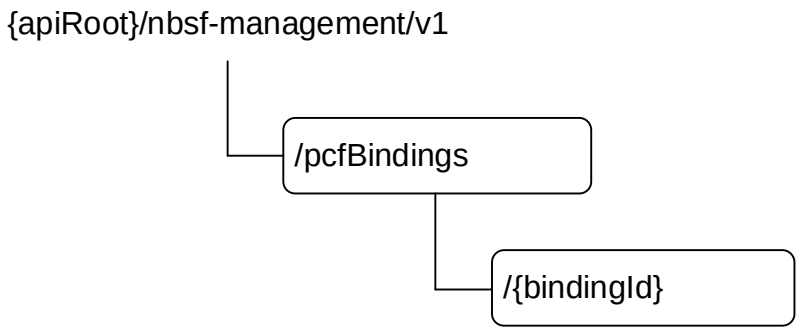


Figure 5.3.1-1: Resource URI structure of the Nbsf_Management API

Table 5.3.1-1 provides an overview of the resources and applicable HTTP methods.

Table 5.3.1-1: Resources and methods overview

Resource name	Resource URI	HTTP method or custom operation	Description
PCF Session Bindings	/pcfBindings	POST	Register a new PCF Session binding information of a given UE address in the BSF.
		GET	Retrieve the Session binding information i.e. PCF address information of a given tuple (UE address, SUPI; GPSI, DNN, S-NSSAI).
Individual PCF Session Binding	/pcfBindings /{bindingId}	DELETE	Deregister an existing PCF Session binding information from the BSF.
		PATCH	Update an existing PCF Session binding information in the BSF.

5.3.2 Resource: PCF Session Bindings

5.3.2.1 Description

This resource represents a collection of the different PCF Session binding information of given UE address(es) registered in the BSF.

5.3.2.2 Resource definition

Resource URI: {apiRoot}/nbsf-management/v1/pcfBindings

This resource shall support the resource URI variables defined in table 5.3.2.2-1.

Table 5.3.2.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See subclause 5.1

5.3.2.3 Resource Standard Methods

5.3.2.3.1 POST

This method shall support the URI query parameters specified in table 5.3.2.3.1-1.

Table 5.3.2.3.1-1: URI query parameters supported by the POST method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.3.2.3.1-2 and the response data structures and response codes specified in table 5.3.2.3.1-3.

Table 5.3.2.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
PcfBinding	M	1	Register a new Individual PCF binding information.

Table 5.3.2.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
PcfBinding	M	1	201 Created	The creation of an individual PCF session biding.
ExtProblemDetails	O	0..1	403 Forbidden	The existing PCF binding information stored in the BSF for the indicated combination is returned.
NOTE: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [6] shall also apply.				

Table 5.3.2.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/nbsf-management/v1/pcfBindings/{bindingId}

5.3.2.3.2 GET

This method shall support the URI query parameters specified in table 5.3.2.3.2-1.

Table 5.3.2.3.2-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
ipv4Addr	Ipv4Addr	C	0..1	The IPv4 Address of the served UE. (NOTE 1) (NOTE 3)
ipv6Prefix	Ipv6Prefix	C	0..1	The IPv6 Address of the served UE. (NOTE 1) (NOTE 3) The NF service consumer shall append '/128' to the IPv6 address in the attribute value. E.g. '2001:db8:85a3::8a2e:370:7334/128'.
macAddr48	MacAddr48	C	0..1	The MAC Address of the served UE. (NOTE 1)
dnn	Dnn	O	0..1	DNN
supi	Supi	O	0..1	Subscription Permanent Identifier
gpsi	Gpsi	O	0..1	Generic Public Subscription Identifier
snssai	Snssai	O	0..1	The identification of slice. (NOTE 2)
ipDomain	string	O	0..1	The IPv4 address domain identifier. (NOTE 2)
supp-feat	SupportedFeatures	O	0..1	To filter irrelevant responses related to unsupported features.

NOTE 1: One and only one of query parameter ipv4Addr, ipv6Prefix or macAddr48 shall be present.
NOTE 2: The query parameters snssai and/or ipDomain, if applicable (IPv4 address overlapping), shall be present with query parameter ipv4Addr.
NOTE 3: 5G-RG and FN-RG replaces UE for wireline access support. See 3GPP TS 23.316 [19].
NOTE 4: The ipv4Addr and ipv6Prefix query parameters may include the IP address of devices in networks behind the UE (see subclause 5.6.14 of 3GPP TS 23.501 [2]).

This method shall support the request data structures specified in table 5.3.2.3.2-2 and the response data structures and response codes specified in table 5.3.2.3.2-3.

Table 5.3.2.3.2-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.3.2.3.2-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
PcfBinding	M	1	200 OK	The individual PCF session binding information resource matching the query parameter(s) is returned.
n/a			204 No Content	There is no PCF session binding information matching the query parameter(s).
ProblemDetails	O	0..1	400 Bad Request	More than one binding information is found. (NOTE 2)

NOTE 1: The mandatory HTTP error status codes for the GET method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [6] shall also apply.
NOTE 2: Failure cases are described in subclause 5.7.

5.3.3 Resource: Individual PCF Session Binding

5.3.3.1 Description

This resource represents a collection of the different PCF Session binding information of given UE address(es) registered in the BSF.

5.3.3.2 Resource definition

Resource URI: **{apiRoot}/nbsf-management/v1/pcfBindings/{bindingId}**

This resource shall support the resource URI variables defined in table 5.3.3.2-1.

Table 5.3.3.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See subclause 5.1
bindingId	string	Represents the individual PCF Session Binding. To enable that the value is used as part of a URI, the string shall only contain characters allowed according to the "lower-with-hyphen" naming convention defined in 3GPP TS 29.501 [7].

5.3.3.3 Resource Standard Methods

5.3.3.3.1 DELETE

This method shall support the URI query parameters specified in table 5.3.3.3.2-1.

Table 5.3.3.3.1-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.3.3.3.1-2 and the response data structures and response codes specified in table 5.3.3.3.1-3.

Table 5.3.3.3.1-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 5.3.3.3.1-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful case: The Individual PCF session binding information resource is deleted.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection, during Individual PCF Session Binding deletion. The response shall include a Location header field containing an alternative URI of the resource located in an alternative BSF (service) instance. Applicable if the feature "ES3XX" is supported.
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection, during Individual PCF Session Binding deletion. The response shall include a Location header field containing an alternative URI of the resource located in an alternative BSF (service) instance. Applicable if the feature "ES3XX" is supported.
NOTE: The mandatory HTTP error status codes for the POST method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [6] shall also apply.				

Table 5.3.3.3.1-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located in an alternative BSF (service) instance.
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance towards which the request is redirected.

Table 5.3.3.3.1-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located in an alternative BSF (service) instance.
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance towards which the request is redirected.

5.3.3.3.2 PATCH

This method shall support the URI query parameters specified in table 5.3.3.3.2-1.

Table 5.3.3.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 5.3.3.3.2-2 and the response data structures and response codes specified in table 5.3.3.3.2-3.

Table 5.3.3.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
PcfBindingPatch	M	1	Update an individual PCF binding information.

Table 5.3.3.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
PcfBinding	M	1	200 OK	Successful case: The Individual PCF session binding information resource is updated.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection, during Individual PCF Session Binding modification. The response shall include a Location header field containing an alternative URI of the resource located in an alternative BSF (service) instance. Applicable if the feature "ES3XX" is supported.
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection, during Individual PCF Session Binding modification. The response shall include a Location header field containing an alternative URI of the resource located in an alternative BSF (service) instance. Applicable if the feature "ES3XX" is supported.
NOTE: The mandatory HTTP error status codes for the PATCH method listed in table 5.2.7.1-1 of 3GPP TS 29.500 [6] shall also apply.				

Table 5.3.3.3.2-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located in an alternative BSF (service) instance.
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance towards which the request is redirected.

Table 5.3.3.3.2-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located in an alternative BSF (service) instance.
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance towards which the request is redirected.

5.4 Custom Operations without associated resources

None in this release of this specification.

5.5 Notifications

None in this release of this specification.

5.6 Data Model

5.6.1 General

This subclause specifies the application data model supported by the API.

Table 5.6.1-1 specifies the data types defined for the Nbsf_Management service based interface protocol.

Table 5.6.1-1: Nbsf_Management specific Data Types

Data type	Section defined	Description	Applicability
BindingResp	5.6.2.6	Contains the binding information.	SamePcf
BindingLevel	5.6.3.3	Contains the binding level.	
ParameterCombination	5.6.2.4	The combination used by the BSF to check whether there is an existing PCF binding information.	SamePcf
ExtProblemDetails	5.6.2.5	Contains the FQDN or IP endpoints of the existing PCF and cause value if there is an existing PCF binding information for the indicated combination.	SamePcf
PcfBinding	5.6.2.2	Identifies an Individual PCF binding.	
PcfBindingPatch	5.6.2.3	Identifies an Individual PCF binding used for Patch method.	BindingUpdate

Table 5.6.1-2 specifies data types re-used by the Nbsf_Management service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nbsf_Management service based interface.

Table 5.6.1-2: Nbsf_Management re-used Data Types

Data type	Reference	Comments	Applicability
DateTime	3GPP TS 29.571 [10]		
DiameterIdentity	3GPP TS 29.571 [10]		
Dnn	3GPP TS 29.571 [10]		
Fqdn	3GPP TS 29.510 [12]		
Gpsi	3GPP TS 29.571 [10]		
IpEndPoint	3GPP TS 29.510 [12]		
Ipv4Addr	3GPP TS 29.571 [10]		
Ipv4AddrMask	3GPP TS 29.571 [11]	String identifying an IPv4 address mask.	
Ipv4AddrRm	3GPP TS 29.571 [10]		
Ipv6Prefix	3GPP TS 29.571 [10]		
Ipv6PrefixRm	3GPP TS 29.571 [10]		
MacAddr48	3GPP TS 29.571 [10]		
MacAddr48Rm	3GPP TS 29.571 [10]		
NfInstanceId	3GPP TS 29.571 [10]		
NfSetId	3GPP TS 29.571 [10]		
ProblemDetails	3GPP TS 29.571 [10]	Used in error responses to provide more detailed information about an error.	
RedirectResponse	3GPP TS 29.571 [10]	Contains redirection related information.	ES3XX
Snssai	3GPP TS 29.571 [10]		
Supi	3GPP TS 29.571 [10]		
SupportedFeatures	3GPP TS 29.571 [10]	Used to negotiate the applicability of the optional features defined in table 5.8-1.	

5.6.2 Structured data types

5.6.2.1 Introduction

This subclause defines the structures to be used in resource representations.

5.6.2.2 Type PcfBinding

Table 5.6.2.2-1: Definition of type PcfBinding

Attribute name	Data type	P	Cardinality	Description	Applicability
supi	Supi	O	0..1	Subscription Permanent Identifier	
gpsi	Gpsi	O	0..1	Generic Public Subscription Identifier	
ipv4Addr	Ipv4Addr	C	0..1	The IPv4 Address of the served UE. (NOTE 4) (NOTE 8)	
ipv6Prefix	Ipv6Prefix	C	0..1	The IPv6 Address Prefix of the served UE. (NOTE 4) (NOTE 5) (NOTE 8)	
addIpv6Prefixes	array(Ipv6Prefix)	O	1..N	The additional IPv6 Address Prefixes of the served UE. (NOTE 4) (NOTE 5) (NOTE 8)	MultiUeAddr
ipDomain	string	O	0..1	IPv4 address domain identifier. (NOTE 1) (NOTE 8)	
macAddr48	MacAddr48	C	0..1	The MAC Address of the served UE. (NOTE 8)	
addMacAddrs	array(MacAddr48)	O	1..N	The additional MAC Addresses of the served UE. (NOTE 8)	MultiUeAddr
dnn	Dnn	M	1	DNN, a full DNN with both the Network Identifier and Operator Identifier, or a DNN with the Network Identifier only.	
pcfFqdn	Fqdn	C	0..1	FQDN of the PCF hosting the Npcf_PolicyAuthorization service. (NOTE 2) (NOTE 9)	
pcfIpEndPoints	array(IpEndPoint)	C	1..N	IP end points of the PCF hosting the Npcf_PolicyAuthorization service. (NOTE 2) (NOTE 9)	
pcfDiamHost	DiameterIdentity	C	0..1	The diameter host for an individual PCF. (NOTE 3) (NOTE 9)	
pcfDiamRealm	DiameterIdentity	C	0..1	The diameter realm for an individual PCF. (NOTE 3) (NOTE 9)	
pcfSmFqdn	Fqdn	O	0..1	FQDN of the PCF hosting the Npcf_SMPolicyControl service. (NOTE 7)	SamePcf
pcfSmIpEndPoints	array(IpEndPoint)	O	1..N	IP end points of the PCF hosting the Npcf_SMPolicyControl service. (NOTE 7)	SamePcf
snssai	Snssai	M	1	The identification of slice.	
suppFeat	SupportedFeatures	C	0..1	Used to negotiate the supported optional features as described in subclause 5.8. Shall be present in the HTTP POST request/response; or in the HTTP GET response if the "supp-feat" attribute query parameter is included in the HTTP GET request.	
pcfId	NfInstanceId	O	0..1	PCF instance identifier	
pcfSetId	NfSetId	O	0..1	The PCF set Id	
recoveryTime	DateTime	O	0..1	Recovery time of the PCF	
paraCom	ParameterCombination	O	0..1	If it is included, the BSF shall check whether there is an existing PCF binding information for the indicated combination. (NOTE 6)	SamePcf
bindLevel	BindingLevel	O	0..1	Contains the level of binding.	
ipv4FrameRouteList	array(Ipv4AddrMask)	O	1..N	List of Framed Route information of IPv4.	
ipv6FrameRouteList	array(Ipv6Prefix)	O	1..N	List of Framed Route information of IPv6.	

NOTE 1: The ipDomain attribute may only be provided if the ipv4Addr attribute is present.
NOTE 2: When the "ExtendedSamePcf" feature is not supported, at least one of "pcfFqdn" or "pcfIpEndPoints" shall be included if the PCF supports the Npcf_PolicyAuthorization service. When the "ExtendedSamePcf" feature is supported these attributes may be provided if available.
NOTE 3: When the "ExtendedSamePcf" feature is not supported, both pcfDiamHost and pcfDiamRealm are provided if the PCF supports Rx interface. When the "ExtendedSamePcf" feature is supported these attributes may provided if available.
NOTE 4: 5G-RG and FN-RG replaces UE for wireline access support. See 3GPP TS 23.316 [19].
NOTE 5: IPv6 prefix(es) shorter than /64 or full IPv6 address(es_ with a /128 prefix may be encoded as the "ipv6Prefix" and "addIpv6Prefixes" attributes, according to 3GPP TS 23.316 [19], subclause 8.3.1.
NOTE 6: If the BSF finds that there is an existing Individual PCF Session binding resource for the indicated combination containing Npcf_SMPolicyControl service addressing information, the BSF shall not check other Individual PCF Session binding resources and shall reject the ongoing registration, and return the FQDN or IP endpoints of the Npcf_SMPolicyControl service of the matching Individual PCF Session binding resource to the requesting PCF.
NOTE 7: At least one of the "pcfSmFqdn" attribute or the "pcfSmIpEndPoints" attribute shall be included in the binding information, if the binding refers to an SM Policy association and if the "SamePcf" feature is supported and the PCF determines that the same PCF shall be selected for the SM Policy associations with the same SUPI/DNN/S-NSSAI parameter combination in the non-roaming or home-routed scenario based on operator's policies and configuration.
NOTE 8: When the "ExtendedSamePcf" feature is not supported the address information of the served UE shall be provided, i.e., either the "ipv4Addr", the "ipv6Prefix" and/or "addIpv6Prefixes" attributes or the "macAddr48" and/or "addMacAddrs" attributes shall be provided as specified in subclause 4.2.2.2.
NOTE 9: When the "ExtendedSamePcf" feature is not supported the address information of the Npcf_PolicyAuthorization service and/or Rx interface shall be provided, i.e., both "pcfDiamHost" and "pcfDiamRealm" and/or at least one of the "pcfFqdn" or "pcfIpEndPoints" shall be provided as specified in subclause 4.2.2.2.

5.6.2.3 Type PcfBindingPatch

Table 5.6.2.3-1: Definition of type PcfBindingPatch

Attribute name	Data type	P	Cardinality	Description	Applicability
ipv4Addr	Ipv4AddrRm	O	0..1	The IPv4 Address of the served UE. (NOTE 2)	
ipDomain	string	O	0..1	IPv4 address domain identifier. (NOTE 1)	
ipv6Prefix	Ipv6PrefixRm	O	0..1	The IPv6 Address Prefix of the served UE. (NOTE 2) (NOTE 3)	
addIpv6Prefixes	array(Ipv6Prefix)	O	1..N	The additional IPv6 Address Prefixes of the served UE. (NOTE 2) (NOTE 3)	MultiUeAddr
macAddr48	MacAddr48Rm	O	0..1	The MAC Address of the served UE.	
addMacAddrs	array(MacAddr48)	O	1..N	The additional MAC Addresses of the served UE.	MultiUeAddr
pcfId	NfInstanceId	O	0..1	PCF instance identifier	
pcfFqdn	Fqdn	O	0..1	FQDN of the PCF hosting the Npcf_PolicyAuthorization service.	
pcfIpEndPoints	array(IpEndPoint)	O	1..N	IP end points of the PCF hosting the Npcf_PolicyAuthorization service.	
pcfDiamHost	DiameterIdentity	O	0..1	The diameter host for an individual PCF.	
pcfDiamRealm	DiameterIdentity	O	0..1	The diameter realm for an individual PCF.	

NOTE 1: If applicable, the consumer (e.g. PCF) shall also request to remove the ipDomain attribute if the ipv4Addr attribute is requested to be removed.
NOTE 2: 5G-RG and FN-RG replaces UE for wireline access support. See 3GPP TS 23.316 [19].
NOTE 3: IPv6 prefix(es) shorter than /64 or full IPv6 address(es) with a /128 prefix can be encoded as the "ipv6Prefix" and "addIpv6Prefixes" attributes, according to 3GPP TS 23.316 [19], subclause 8.3.1.

5.6.2.4 Type ParameterCombination

Table 5.6.2.4-1: Definition of type ParameterCombination

Attribute name	Data type	P	Cardinality	Description	Applicability
supi	Supi	O	0..1	Subscription Permanent Identifier	
dnn	Dnn	O	0..1	DNN, a full DNN with both the Network Identifier and Operator Identifier, or a DNN with the Network Identifier only.	
snssai	Snssai	O	0..1	The identification of slice.	
NOTE 1: At least one of the attributes in this table shall be included. NOTE 2: The applicable parameter combinations in a given deployment shall be disjoint combinations. E.g., if a deployment requires a parameter combination that includes a SUPI value for a DNN/S-NSSAI combination, subsequent parameter combinations of that DNN/S-NSSAI combination shall also include the corresponding SUPI attribute.					

5.6.2.5 Type ExtProblemDetails

Table 5.6.2.5-1: Definition of type ExtProblemDetails as a list of to be combined data types

Data Type	P	Cardinality	Description	Applicability
ProblemDetails	O	0..1	Problem Details	
BindingResp	O	0..1	PCF Binding Information	

5.6.2.6 Type BindingResp

Table 5.6.2.6-1: Definition of type BindingResp

Attribute name	Data type	P	Cardinality	Description	Applicability
pcfSmFqdn	Fqdn	O	0..1	FQDN of the PCF hosting the Npcf_SMPolicyControl service. (NOTE)	
pcfSmIpEndPoints	array(IpEndPoint)	O	1..N	IP end points of the PCF hosting the Npcf_SMPolicyControl service. (NOTE)	
NOTE: Either the "pcfSmFqdn" attribute or the "pcfSmIpEndPoints" attribute shall be included.					

5.6.3 Simple data types and enumerations

5.6.3.1 Introduction

This subclause defines simple data types and enumerations that can be referenced from data structures defined in the previous subclauses.

5.6.3.2 Simple data types

The simple data types defined in table 5.6.3.2-1 shall be supported.

Table 5.6.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability
n/a			

5.6.3.3 Enumeration: BindingLevel

Table 5.6.3.3-1: Enumeration BindingLevel

Enumeration value	Description	Applicability
NF_SET	Indicates the NF set level of binding.	
NF_INSTANCE	Indicates the NF instance level of binding.	

5.7 Error handling

5.7.1 General

HTTP error handling shall be supported as specified in subclause 5.2.4 of 3GPP TS 29.500 [6].

For the Nbsf_Management Service API, HTTP error responses shall be supported as specified in subclause 4.8 of 3GPP TS 29.501 [7]. Protocol errors and application errors specified in table 5.2.7.2-1 of 3GPP TS 29.500 [6] shall be supported for an HTTP method if the corresponding HTTP status codes are specified as mandatory for that HTTP method in table 5.2.7.1-1 of 3GPP TS 29.500 [6].

In addition, the requirements in the following subclauses shall apply.

5.7.2 Protocol Errors

In this Release of the specification, there are no additional protocol errors applicable for the Nbsf_Management Service API.

5.7.3 Application Errors

The application errors defined for the Nbsf_Management Service API are listed in table 5.7.3-1.

Table 5.7.3-1: Application errors

Application Error	HTTP status code	Description
MULTIPLE_BINDING_INFO_FOUND	400 Bad Request	Indicates that the BSF found more than one binding resource so it cannot provide the selected PCF to the consumer. (NOTE 1)
EXISTING_BINDING_INFO_FOUND	403 Forbidden	Indicates that the BSF found an existing PCF binding information for the indicated combination. (NOTE 2)
NOTE 1: This application error is included in the responses to the GET request. NOTE 2: This application error is included in the responses to the POST request. NOTE 3: Including a "ProblemDetails" data structure with the "cause" attribute in the HTTP response is optional unless explicitly mandated in the service operation subclauses.		

5.8 Feature negotiation

The optional features in table 5.8-1 are defined for the Nbsf_Management Service API. They shall be negotiated using the extensibility mechanism defined in subclause 6.6 of 3GPP TS 29.500 [6].

Table 5.8-1: Supported Features

Feature number	Feature Name	Description
1	MultiUeAddr	This feature indicates the support of multiple UE addresses (IPv6 prefixes or MAC addresses) in the same binding information.
2	BindingUpdate	The consumer can use this feature for updating the session binding information.
3	SamePcf	This feature indicates the support of same PCF selection for the indicated combination. (NOTE)
4	ES3XX	Extended Support for 3xx redirections. This feature indicates the support of redirection for any service operation, according to Stateless NF procedures as specified in subclauses 6.5.3.2 and 6.5.3.3 of 3GPP TS 29.500 [6] and according to HTTP redirection principles for indirect communication, as specified in subclause 6.10.9 of 3GPP TS 29.500 [6].
5	ExtendedSamePcf	This feature extends the support of same PCF selection for the indicated combination. This feature requires the support of SamePcf feature. (NOTE)
NOTE: The "SamePcf" feature is applicable to the deployments where the N5 and/or Rx interface apply and the UE address is available in the PCF at the creation of the SM Policy Association. The "ExtendedSamePcf" feature is applicable for any PCF deployment, regardless of UE address availability at the creation of SM Policy association and/or N5 and/or Rx applicability.		

5.9 Security

As indicated in 3GPP TS 33.501 [15] and 3GPP TS 29.500 [6], the access to the Nbsf_Management API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [16]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see 3GPP TS 29.510 [12]) plays the role of the authorization server.

If OAuth2 is used, a n NF Service Consumer, prior to consuming services offered by the Nbsf_Management API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in 3GPP TS 29.510 [12], subclause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nbsf_Management service.

The Nbsf_Management API defines a single scope "nbsf-management" for the entire service, and it does not define any additional scopes at resource or operation level.

Annex A (normative): OpenAPI specification

A.1 General

The present Annex contains an OpenAPI [11] specification of HTTP messages and content bodies used by the Nbsf_Management API.

This Annex shall take precedence when being discrepant to other parts of the specification with respect to the encoding of information elements and methods within the API.

NOTE: The semantics and procedures, as well as conditions, e.g. for the applicability and allowed combinations of attributes or values, not expressed in the OpenAPI definitions but defined in other parts of the specification also apply.

Informative copies of the OpenAPI specification file contained in this 3GPP Technical Specification are available on a Git-based repository, that uses the GitLab software version control system (see clause 5B of the 3GPP TR 21.900 [18] and subclause 5.3.1 of the 3GPP TS 29.501 [7] for further information).

A.2 Nbsf_Management API

```
openapi: 3.0.0
info:
  version: 1.1.2
  title: Nbsf_Management
  description: |
    Binding Support Management Service API.
    © 2021, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.
externalDocs:
  description: 3GPP TS 29.521 V16.8.0; 5G System; Binding Support Management Service.
  url: 'http://www.3gpp.org/ftp/Specs/archive/29_series/29.521/'
servers:
  - url: '{apiRoot}/nbsf-management/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in subclause 4.4 of 3GPP TS 29.501.
security:
  - {}
  - oAuth2ClientCredentials:
    - nbsf-management
paths:
  /pcfBindings:
    post:
      summary: Create a new Individual PCF binding information
      operationId: CreatePCFBinding
      tags:
        - PCF Bindings (Collection)
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/PcfBinding'
      responses:
        '201':
          description: The creation of an individual PCF session binding.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/PcfBinding'
          headers:
            Location:
              description: 'Contains the URI of the newly created resource, according to the
structure: {apiRoot}/nbsf-management/v1/pcfBindings/{bindingId}'
              required: true
```

```
              schema:
                type: string
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          description: The existing PCF binding information stored in the BSF for the indicated
combination is returned.
          content:
            application/problem+json:
              schema:
                $ref: '#/components/schemas/ExtProblemDetails'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    get:
      summary: Read PCF Bindings information
      operationId: GetPCFBindings
      tags:
        - PCF Bindings (Collection)
      parameters:
        - name: ipv4Addr
          in: query
          description: The IPv4 Address of the served UE.
          required: false
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
        - name: ipv6Prefix
          in: query
          description: The IPv6 Address of the served UE. The NF service consumer shall append
'/128' to the IPv6 address in the attribute value. E.g. '2001:db8:85a3::8a2e:370:7334/128'.
          required: false
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Prefix'
        - name: macAddr48
          in: query
          description: The MAC Address of the served UE.
          required: false
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
        - name: dnn
          in: query
          description: DNN.
          required: false
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
        - name: supi
          in: query
          description: Subscription Permanent Identifier.
          required: false
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        - name: gpsi
          in: query
          description: Generic Public Subscription Identifier
          required: false
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
        - name: snssai
          in: query
          description: The identification of slice.
          required: false
          content:
            application/json:
```

```
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
        - name: ipDomain
          in: query
          description: The IPv4 address domain identifier.
          required: false
          schema:
            type: string
        - name: supp-feat
          in: query
          description: To filter irrelevant responses related to unsupported features
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      responses:
        '200':
          description: The individual PCF session binding session binding information resource
matching the query parameter(s) is returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/PcfBinding'
        '204':
          description: There is no PCF session binding information matching the query parameter(s).
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '414':
          $ref: 'TS29571_CommonData.yaml#/components/responses/414'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  /pcfBindings/{bindingId}:
    delete:
      summary: Delete an existing Individual PCF Binding information
      operationId: DeleteIndPCFBinding
      tags:
        - Individual PCF Binding (Document)
      parameters:
        - name: bindingId
          in: path
          description: Represents the individual PCF Session Binding.
          required: true
          schema:
            type: string
      responses:
        '204':
          description: No Content. The Individual PCF session binding information resource is
deleted.
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
```

```
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    patch:
      summary: Update an existing Individual PCF Binding information
      operationId: UpdateIndPCFBinding
      tags:
        - Individual PCF Binding (Document)
      parameters:
        - name: bindingId
          in: path
          description: Represents the individual PCF Session Binding.
          required: true
          schema:
            type: string
      requestBody:
        description: Parameters to update the existing session binding
        required: true
        content:
          application/merge-patch+json:
            schema:
              $ref: '#/components/schemas/PcfBindingPatch'
      responses:
        '200':
          description: OK (Successful update of the session binding)
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/PcfBinding'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'
components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            nbsf-management: Access to the Nbsf_Management API
  schemas:
    PcfBinding:
      type: object
      properties:
        supi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
        gpsi:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
        ipv4Addr:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4Addr'
        ipv6Prefix:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Prefix'
        addIpv6Prefixes:
          type: array
          items:
```

```
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Prefix'
        minItems: 1
        description: The additional IPv6 Address Prefixes of the served UE.
      ipDomain:
        type: string
      macAddr48:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
      addMacAddrs:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
        minItems: 1
        description: The additional MAC Addresses of the served UE.
      dnn:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      pcfFqdn:
        $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/Fqdn'
      pcfIpEndPoints:
        type: array
        items:
          $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/IpEndPoint'
        minItems: 1
        description: IP end points of the PCF hosting the Npcf_PolicyAuthorization service.
      pcfDiamHost:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DiameterIdentity'
      pcfDiamRealm:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DiameterIdentity'
      pcfSmFqdn:
        $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/Fqdn'
      pcfSmIpEndPoints:
        type: array
        items:
          $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/IpEndPoint'
        minItems: 1
        description: IP end points of the PCF hosting the Npcf_SMPolicyControl service.
      snssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
      suppFeat:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
      pcfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
      pcfSetId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfSetId'
      recoveryTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      paraCom:
        $ref: '#/components/schemas/ParameterCombination'
      bindLevel:
        $ref: '#/components/schemas/BindingLevel'
      ipv4FrameRouteList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4AddrMask'
        minItems: 1
      ipv6FrameRouteList:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Prefix'
        minItems: 1
    required:
      - dnn
      - snssai
  PcfBindingPatch:
    type: object
    properties:
      ipv4Addr:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv4AddrRm'
      ipDomain:
        type: string
        nullable: true
      ipv6Prefix:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6PrefixRm'
      addIpv6Prefixes:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Ipv6Prefix'
        minItems: 1
        description: The additional IPv6 Address Prefixes of the served UE.
```

```
        nullable: true
      macAddr48:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48Rm'
      addMacAddrs:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/MacAddr48'
        minItems: 1
        description: The additional MAC Addresses of the served UE.
        nullable: true
      pcfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
      pcfFqdn:
        $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/Fqdn'
      pcfIpEndPoints:
        type: array
        items:
          $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/IpEndPoint'
        minItems: 1
        description: IP end points of the PCF hosting the Npcf_PolicyAuthorization service.
      pcfDiamHost:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DiameterIdentity'
      pcfDiamRealm:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DiameterIdentity'
  ParameterCombination:
    type: object
    properties:
      supi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
      dnn:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Dnn'
      snssai:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Snssai'
  ExtProblemDetails:
    allOf:
      - $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
      - $ref: '#/components/schemas/BindingResp'
  BindingResp:
    type: object
    properties:
      pcfSmFqdn:
        $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/Fqdn'
      pcfSmIpEndPoints:
        type: array
        items:
          $ref: 'TS29510_Nnrf_NFManagement.yaml#/components/schemas/IpEndPoint'
        minItems: 1
        description: IP end points of the PCF hosting the Npcf_SMPolicyControl service.
  BindingLevel:
    anyOf:
    - type: string
      enum:
        - NF_SET
        - NF_INSTANCE
    - type: string
      description: >
        This string provides forward-compatibility with future
        extensions to the enumeration but is not used to encode
        content defined in the present version of this API.
    description: >
      Possible values are
      - "NF_SET"
      - "NF_INSTANCE"
```

Annex B (informative): Deployment option to support BSF and DRA coexistence due to network migration

As described in Annex B of 3GPP TS 23.503 [4], the Diameter Routing Agent (DRA) and the BSF can coexist in an operator's network during the network migration to 5GC. The DRA is described in 3GPP TS 29.213 [14] and can be a service consumer of the Nbsf_Management service.

During the Rx session establishment, the DRA can discover the selected PCF for the related subscriber by using the Nbsf_Management_Discovery service operation to obtain the related PCF address if it has no stored binding information derived from an ongoing Gx session for that subscriber.

NOTE 1: For a UE in the EPC there is a Gx session and the DRA stores the binding information. For a UE in the 5GC the Npcf_SmPolicyControl service is used and the BSF stores the binding information.

NOTE 2: If the DRA has no stored binding information derived from an ongoing Gx session for a subscriber, the DRA needs to request new binding information for each Rx session establishment because the information in the BSF could have changed compared to any previous binding information the DRA requested.

Annex C (informative): Change history

Change history							
Date	TSG #	TSG Doc.	CR	Rev	Cat	Subject/Comment	New
2018-01						TS skeleton of Binding Support Management Service specification	0.0.0
2018-01						Inclusion of documents agreed in CT3#94 C3-180301, C3-180191, C3-180192 and C3-180193.	0.1.0
2018-03						Inclusion of documents agreed in CT3#95 C3-181350 and C3-181352.	0.2.0
2018-04						Inclusion of documents agreed in CT3#96 C3-182424 and C3-182510.	0.3.0
2018-05						Inclusion of documents agreed in CT3#97 C3-183287, C3-183500, C3-183881, C3-183502 and C3-183733.	0.4.0
2018-06	CT#80	CP-181031				TS sent to plenary for approval.	1.0.0
2018-06	CT#80	CP-181031				TS approved by plenary	15.0.0
2018-09	CT#81	CP-182015	0001	2	F	PCF id correction for BSF	15.1.0
2018-09	CT#81	CP-182015	0002		F	Reference update: OpenAPI specification	15.1.0
2018-09	CT#81	CP-182015	0004	2	F	Clarification on mandatory HTTP error status codes	15.1.0
2018-09	CT#81	CP-182015	0005	6	B	OpenAPI for TS 29.521	15.1.0
2018-09	CT#81	CP-182015	0006	1	F	Description of Structured data types	15.1.0
2018-09	CT#81	CP-182015	0007	1	B	Support of IPv4 overlapping	15.1.0
2018-09	CT#81	CP-182015	0008		F	Correction of the service name	15.1.0
2018-09	CT#81	CP-182015	0009	1	F	Resource structure presentation	15.1.0
2018-12	CT#82	CP-183205	0011		F	Default value for apiRoot Default value for apiRoot	15.2.0
2018-12	CT#82	CP-183205	0012		F	Correction to DELETE Method for Nbsf_Management Service API	15.2.0
2018-12	CT#82	CP-183205	0013		F	Correction to Typos in URI Paths	15.2.0
2018-12	CT#82	CP-183205	0015		F	API version	15.2.0
2018-12	CT#82	CP-183205	0016		F	ExternalDocs OpenAPI field	15.2.0
2018-12	CT#82	CP-183205	0017		F	Location header field in OpenAPI	15.2.0
2018-12	CT#82	CP-183205	0018	1	F	Security	15.2.0
2018-12	CT#82	CP-183205	0019	1	F	supported content types	15.2.0
2018-12	CT#82	CP-183205	0020	2	F	HTTP Error responses	15.2.0
2018-12	CT#82	CP-183205	0021	2	F	DRA as service consumer	15.2.0
2018-12	CT#82	CP-183205	0023		F	Change presence in BSF binding	15.2.0
2018-12	CT#82	CP-183205	0024	1	F	Presence conditions in yaml file	15.2.0
2018-12	CT#82	CP-183205	0025	1	F	Missing 201 response body for POST to /pcfBindings	15.2.0
2019-03	CT#83	CP-190113	0028	2	F	Handling of unsupported query parameter combinations	15.3.0
2019-03	CT#83	CP-190113	0029	1	F	Correction of description of the Nbsf_Management_Register Service and Nbsf_Management_Discovery service operations	15.3.0
2019-03	CT#83	CP-190113	0030		F	BSF resource cleanup	15.3.0
2019-03	CT#83	CP-190113	0031	1	F	Formatting of structured data types in query parameters	15.3.0
2019-03	CT#83	CP-190113	0032	1	F	Correction on the handling of UE addresses	15.3.0
2019-03	CT#83	CP-190110	0033	2	F	Miscellaneous BSF correction	15.3.0
2019-03	CT#83	CP-190140	0034	1	F	OpenAPI Version number update	15.3.0
2019-06	CT#84	CP-191079	0036		F	Remove NSI ID	15.4.0
2019-06	CT#84	CP-191106	0037	5	B	Support multiple UE addresses in one binding	16.0.0
2019-06	CT#84	CP-191106	0038	5	B	Binding update support	16.0.0
2019-06	CT#84	CP-191079	0039	1	F	Precedence of OpenAPI file	15.4.0
2019-06	CT#84	CP-191079	0040	1	F	Copyright Note in YAML files	15.4.0
2019-06	CT#84	CP-191089	0041	1	F	Correction of Location header in Nbsf_Management OpenAPI	16.0.0
2019-06	CT#84	CP-191101	0043	2	F	OpenAPI version number update	16.0.0
2019-09	CT#85	CP-192199	0045	2	F	Session binding for IPv6 addresses	16.1.0
2019-09	CT#85	CP-192156	0046		F	Support multiple UE addresses in BSF	16.1.0
2019-09	CT#85	CP-192152	0047	1	B	IP address handling in wireline access	16.1.0
2019-09	CT#85	CP-192234	0050	2	F	OpenAPI version update TS 29.521 Rel-16	16.1.0
2019-12	CT#86	CP-193197	0053	3	B	Same PCF selection for the same UE ID, S-NSSAI and DNN combination	16.2.0
2019-12	CT#86	CP-193197	0054		F	Update of API version and TS version in OpenAPI file	16.2.0
2020-03	CT#87e	CP-200207	0055	1	B	Update of the same PCF selection	16.3.0
2020-03	CT#87e	CP-200207	0056		B	DNN Clarification	16.3.0
2020-03	CT#87e	CP-200208	0058	2	B	Adding NWDAF as Nbsf_management service consumer	16.3.0
2020-03	CT#87e	CP-200207	0059		F	Resolve editor note for PATCH	16.3.0
2020-03	CT#87e	CP-200207	0060	1	F	Miscellaneous errors	16.3.0
2020-03	CT#87e	CP-200253	0061	1	F	Support of the Update service operation	16.3.0
2020-03	CT#87e	CP-200214	0062		F	OpenAPI: usage of the "tags" keyword	16.3.0
2020-03	CT#87e	CP-200260	0063	1	B	PCF set Id/PCF Id in Nbsf_Management_Register/Update	16.3.0
2020-03	CT#87e	CP-200215	0064		F	Correction on PcfBinding	16.3.0

2020-03	CT#87e	CP-200216	0065		F	Update of OpenAPI version and TS version in externalDocs field	16.3.0
2020-06	CT#88e	CP-201233	0066	1	F	Corrections on SamePcf	16.4.0
2020-06	CT#88e	CP-201246	0067	1	F	Corrections related to UEaddr	16.4.0
2020-06	CT#88e	CP-201259	0068	3	B	Update of PCF address(es)	16.4.0
2020-06	CT#88e	CP-201275	0069	2	B	Clarification of the DS-TT MAC address	16.4.0
2020-06	CT#88e	CP-201228	0070	3	B	Support of full Frame Routing feature	16.4.0
2020-06	CT#88e	CP-201212	0071	1	F	Binding information retrieval: PCF set ID and PCF instance ID	16.4.0
2020-06	CT#88e	CP-201296	0073	2	F	Correct use of application error	16.4.0
2020-06	CT#88e	CP-201228	0074	1	F	Correct IPv6 prefix	16.4.0
2020-06	CT#88e	CP-201244	0076	1	F	Storage of YAML files in ETSI Forge	16.4.0
2020-06	CT#88e	CP-201246	0080	1	F	Adding DRA as Nbsf_management service consumer	16.4.0
2020-06	CT#88e	CP-201258	0081	1	B	Update of PCF address(es)	16.4.0
2020-06	CT#88e	CP-201256	0083	1	F	URI of the Nbsf_Management service	16.4.0
2020-06	CT#88e	CP-201222	0085	1	A	Correction to the condition of BSF service operations	16.4.0
2020-06	CT#88e	CP-201244	0086	1	F	Optionality of ProblemDetails	16.4.0
2020-06	CT#88e	CP-201233	0087	1	F	suppFeat attribute within PcfBinding data	16.4.0
2020-06	CT#88e	CP-201244	0088	1	F	Supported headers, Resource Data type and yaml mapping	16.4.0
2020-06	CT#88e	CP-201255	0090		F	Update of OpenAPI version and TS version in externalDocs field	16.4.0
2020-09	CT#89e	CP-202077	0092		F	Data type corrections	16.5.0
2020-12	CT#90e	CP-203139	0093	1	F	Essential Corrections and alignments	16.6.0
2021-03	CT#91e	CP-210191	0094	1	F	Support of stateless NFs	16.7.0
2021-03	CT#91e	CP-210202	0095		F	Correction to Framed Routing feature	16.7.0
2021-03	CT#91e	CP-210217	0096		F	Storage of YAML files in ETSI Forge	16.7.0
2021-03	CT#91e	CP-210205	0098	1	F	Correction to SamePcf feature	16.7.0
2021-03	CT#91e	CP-210239	0101		F	Update of OpenAPI version and TS version in externalDocs field	16.7.0
2021-06	CT#92e	CP-211219	0104		F	Correction to Overview and Introduction	16.8.0
2021-06	CT#92e	CP-211200	0106	1	F	Redirect responses with "application/json" media type	16.8.0
2021-06	CT#92e	CP-211219	0108	1	F	Correction to ExtendedSamePcf feature	16.8.0
2021-06	CT#92e	CP-211264	0110		F	Update of OpenAPI version and TS version in externalDocs field	16.8.0
2021-12	CT#94e	CP-213224	0131		F	Correction to PCF Session binding update procedure	16.9.0
2022-03	CT#95e	CP-220176	0141	1	F	Alignment of "Application Errors" clause with SBI TS template	16.10.0
2026-09	CT#113	CP-262127	0238		F	Wrong attribute name for PCF binding	16.11.0

History

Version	Date	Status
V16.4.0	August 2020	Publication
V16.5.0	November 2020	Publication
V16.6.0	January 2021	Publication
V16.7.0	April 2021	Publication
V16.8.0	August 2021	Publication
V16.9.0	January 2022	Publication
V16.10.0	March 2022	Publication
V16.11.0	October 2026	Publication